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DEPARTMENT OF COMPUTER, CONTROL AND MANAGEMENT

Second Assignment Machine Learning

ENGINEERING IN COMPUTER SCIENCE AND ARTIFICIAL
INTELLIGENCE

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Contents

1	Introduction and Motivation	2
1.1	Dataset Selection	2
2	Dataset Description	3
2.1	Regression: Diamonds Dataset	3
2.2	Clustering: Wine Quality (White)	3
2.3	Classification: GTZAN Genre Collection (Image-Based)	4
3	Methodology and Models	5
3.1	Experimental Design: The Three-Pronged Approach	5
3.2	Model Architectures	5
3.3	Dimensionality Reduction Setup	6
3.4	Theoretical Assumptions and Expectations	6
3.5	Evaluation Metrics	7
3.6	Visual Analysis Strategy	7
4	Results and Analysis	8
4.1	Performance Overview	8
4.2	Bias-Variance Analysis (Overfitting vs. Underfitting)	8
4.3	Error Analysis (Confusion Matrices)	9
4.4	Latent Space Analysis (Clustering)	10
4.5	Computational Efficiency vs. Accuracy Trade-off	10
5	Conclusions	11
5.1	Key Findings and Limitations	11
5.2	The Fundamental Challenge	11
5.3	Future Directions	12
5.4	Concluding Remarks	12
	References	13

1 Introduction and Motivation

In the field of Machine Learning, high-dimensional data often presents significant challenges, ranging from increased computational costs to the risk of overfitting—a phenomenon commonly referred to as the "curse of dimensionality" [1]. Consequently, dimensionality reduction has become a cornerstone of modern data preprocessing, aiming to preserve the most relevant information while discarding noise.

The primary objective of this report is to conduct a comparative analysis of three fundamental machine learning tasks: **Regression**, **Clustering**, and **Classification**. We investigate the trade-offs between model performance and data compression by evaluating three distinct experimental setups:

1. **Original Data:** Training models on the full feature set.
2. **PCA Reduction:** Using Principal Component Analysis [2] to project data into a linear subspace.
3. **Autoencoder Reduction:** Utilizing neural networks to learn a non-linear compressed representation [1].

1.1 Dataset Selection

Three diverse datasets were selected to ensure robustness:

- **Regression:** The *Diamonds* dataset [3] to predict prices based on physical attributes.
- **Clustering:** The *Wine Quality* dataset [4] for unsupervised grouping of physicochemical properties.
- **Classification:** The *GTZAN* dataset [5] for music genre recognition via spectrograms.

2 Dataset Description

In this study, we employed three distinct datasets to cover the spectrum of regression, clustering, and classification tasks. All datasets exceed the minimum requirement of 1,000 samples and were preprocessed to ensure compatibility with the dimensionality reduction algorithms.

2.1 Regression: Diamonds Dataset

For the regression task, we utilized the **Diamonds** dataset [3], sourced from the Seaborn library (originally from Kaggle). The objective is to predict the price of a diamond based on its physical attributes.

- **Size:** 53,940 samples.
- **Target Variable:** Price (Continuous value in USD).
- **Features:** The dataset contains 9 independent variables, including continuous measures (Carat, Depth, Table, X, Y, Z dimensions) and categorical attributes (Cut, Color, Clarity).
- **Preprocessing:** To prepare the data for the Feedforward Neural Network (FNN), we applied:
 1. **Label Encoding:** Categorical variables were converted into ordinal integers (e.g., Fair $\rightarrow$ 0, Ideal $\rightarrow$ 4).
 2. **Standardization:** All features were scaled using Z-score normalization (StandardScaler) to facilitate the convergence of gradient descent and PCA [2].

2.2 Clustering: Wine Quality (White)

For the unsupervised clustering task, we selected the **Wine Quality - White** dataset [4] from the UCI Machine Learning Repository. This dataset contains physicochemical data for 4,898 Portuguese "Vinho Verde" wines.

- **Size:** 4,898 samples.
- **Features:** 11 numerical input variables (Fixed acidity, Volatile acidity, Citric acid, Residual sugar, Chlorides, Free sulfur dioxide, Total sulfur dioxide, Density, pH, Sulphates, Alcohol).
- **Task Setup:** The "Quality" label (score between 3 and 9) was removed during training to perform unsupervised K-Means clustering. It was later used only for validation purposes.

- **Preprocessing:** Given that K-Means is distance-based, all features were standardized to have zero mean and unit variance [6].

2.3 Classification: GTZAN Genre Collection (Image-Based)

For the classification task, we utilized the **GTZAN Genre Collection** [5], a benchmark for music genre recognition. To satisfy the assignment requirement for image-based classification, we utilized the Mel-Spectrogram representations of the audio tracks.

- **Size:** 1,000 samples (100 tracks for each of the 10 genres).
- **Classes:** Blues, Classical, Country, Disco, Hiphop, Jazz, Metal, Pop, Reggae, Rock.
- **Input Format:** Grayscale spectrogram images.
- **Preprocessing:**
 1. **Resizing:** Images were resized to 64×64 pixels to reduce the computational load while preserving spectral patterns.
 2. **Normalization:** Pixel values were normalized to the range $[0, 1]$ to assist neural network convergence [1].
 3. **Flattening:** For PCA and Autoencoder training, images were flattened into vectors of size 4,096 (64×64), while the Convolutional Neural Network (CNN) utilized the original 2D structure.

3 Methodology and Models

With the datasets preprocessed as described in Section 2, we proceeded to the model implementation and comparative evaluation phase. This section details the experimental setup, the dimensionality reduction strategies, and the specific configurations for each algorithm. The information in this section is derived from the experimental code implementation and standard machine learning literature [1].

3.1 Experimental Design: The Three-Pronged Approach

A robust comparative strategy was necessary for this study to isolate the effect of dimensionality reduction on model performance. Unlike standard comparative studies that vary only the model architecture, here we varied the **data representation**.

For each of the three tasks (Regression, Clustering, Classification), we executed a strict loop of three experiments:

1. **Baseline (Original Data):** Training the model on the full, high-dimensional feature set to establish a performance upper bound.
2. **Linear Reduction (PCA):** Training the model on data projected into a lower-dimensional subspace using Principal Component Analysis [2].
3. **Non-Linear Reduction (Autoencoder):** Training the model on the latent space representation learned by a neural network-based Autoencoder [1].

All experiments utilized a consistent **80/20 Train-Test split**. For the classification task, this split was **stratified** to maintain class balance across the ten genres.

3.2 Model Architectures

In accordance with the project requirements, we implemented specific algorithms tailored to each task using the Python libraries `scikit-learn` [7] and `TensorFlow/Keras` [8].

- **Feedforward Neural Network (FNN) for Regression:** Chosen to predict diamond prices. The network consists of two dense hidden layers (64 and 32 neurons) with ReLU activation, followed by a linear output layer. This architecture allows the model to capture non-linear relationships between physical attributes and price [1].
- **K-Means for Clustering:** Selected as the unsupervised baseline for the Wine Quality dataset. We fixed $K = 6$ to approximate the distinct quality scores present in the data. The algorithm uses Euclidean distance to iteratively refine cluster centroids [6].

- **Convolutional Neural Network (CNN) for Classification:** Used exclusively for the **Original Image Data**. The architecture features two convolutional blocks (Conv2D + MaxPool) followed by a dense classification head. This allows the model to extract spatial features (time-frequency patterns) from the spectrograms [9].
- **Dense Classifier (MLP) for Reduced Classification:** Since PCA and Autoencoders output flattened 1D vectors, a standard CNN cannot be applied. Therefore, for the reduced datasets, we substituted the CNN with a Multi-Layer Perceptron (MLP) to perform classification on the latent vectors.

3.3 Dimensionality Reduction Setup

To ensure a fair comparison, the target dimensionality (Latent Size) was fixed for both PCA and Autoencoder within each task.

- **PCA Configuration:** We utilized the standard implementation to project data onto the principal orthogonal components that explain the maximum variance [2].
- **Autoencoder Configuration:** We designed a symmetric Encoder-Decoder architecture trained to minimize the Mean Squared Error (MSE) between the input and the reconstruction, optimized via Stochastic Gradient Descent (Adam) [10].
 - *Diamonds*: Reduced from 9 features to **3 latent dimensions**.
 - *Wine Quality*: Reduced from 11 features to **2 latent dimensions** (to facilitate 2D visualization).
 - *GTZAN*: Reduced from 4,096 pixels (64×64) to **128 latent dimensions**.

3.4 Theoretical Assumptions and Expectations

Before analyzing the empirical results, it is crucial to outline the theoretical assumptions.

- **Linear vs. Non-Linear Manifolds:** PCA assumes that the data lies on a linear hyperplane [2]. If the underlying data structure is highly non-linear (likely in image data), PCA is expected to lose significant information compared to the Autoencoder.
- **Spatial Locality Loss:** For the GTZAN classification task, we flatten the 2D spectrograms into vectors before applying PCA or the Autoencoder. We hypothesize that this destruction of spatial structure (time vs. frequency axes) will cause a significant performance drop compared to the CNN.
- **Clustering Geometry:** K-Means assumes spherical clusters. We expect PCA to facilitate this by orthogonalizing features, whereas the Autoencoder might create complex, non-convex manifolds that confuse the simple Euclidean distance metric of K-Means.

3.5 Evaluation Metrics

Models were evaluated on the held-out test set using task-specific metrics:

- **Regression: RMSE** (Root Mean Squared Error) and R^2 **Score** to measure predictive precision.
- **Clustering: Silhouette Score** [11] to evaluate the separation and cohesion of the formed clusters without using ground truth labels.
- **Classification: Accuracy** and **Confusion Matrices** to visualize specific misclassifications between genres.

3.6 Visual Analysis Strategy

While numerical metrics provide a summary, they obscure the dynamics of training. Therefore, a central component of our methodology is the analysis of **Learning Curves** and **Latent Space Projections**.

For the Neural Networks (CNN/FNN), we analyze the divergence between training and validation loss to detect **Overfitting**. For clustering, we plot the 2D projections of the data points to visually verify if the reduction techniques have successfully separated the semantic groups in the latent space.

4 Results and Analysis

This section analyzes the experimental results across the three tasks, focusing on the trade-off between dimensionality reduction and model performance, the generalization capabilities, and the interpretability of the latent spaces.

4.1 Performance Overview

Table 1 details the comparative performance metrics. A clear pattern emerges: models trained on **Original Data** consistently outperform those using reduced representations.

In the classification task, the **CNN (Original)** emerged as the superior model (Accuracy: **61.0%**), significantly outperforming the dense classifiers trained on PCA (46.5%) and Autoencoder (30.0%) features. This significant gap confirms that flattening spectrograms for dimensionality reduction destroys critical **spatial locality** (time-frequency patterns) that convolutional networks exploit [9].

Surprisingly, across both regression and classification, the linear **PCA** technique outperformed the non-linear **Autoencoder**. This suggests that for these specific dataset sizes, the Autoencoder struggled to learn a meaningful manifold, acting closer to a noisy compressor than a feature extractor.

Table 1: Summary of Experimental Results across Tasks

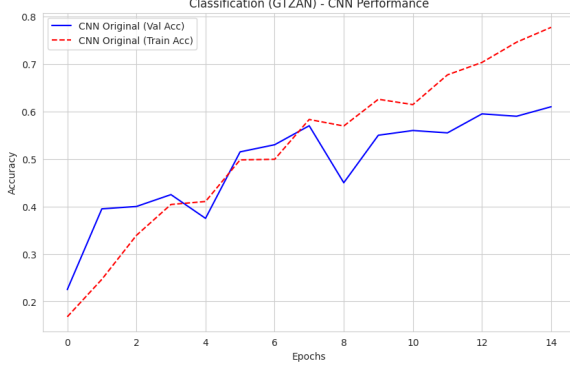
Task	Metric	Original	PCA Reduced	AE Reduced
Regression (Diamonds)	R^2 Score	0.9790	0.8451	0.8338
	RMSE (\$)	577.76	1569.12	1625.20
Clustering (Wine)	Silhouette	<i>Baseline</i>	High Separation	Low Separation
Classification (GTZAN)	Accuracy	0.6100	0.4650	0.3000

4.2 Bias-Variance Analysis (Overfitting vs. Underfitting)

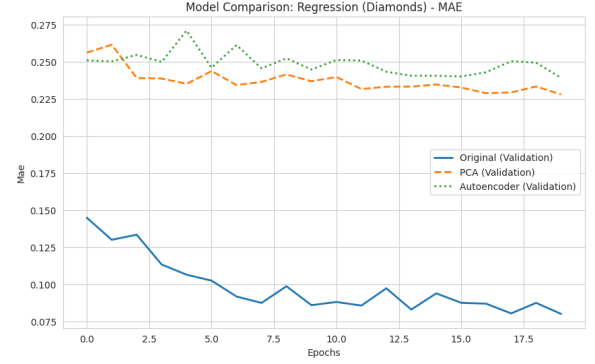
The learning curves reveal two distinct pathological behaviors introduced by the different experimental setups.

1. **CNN Overfitting (Original Data):** As shown in Figure 1a, the CNN exhibits classic **high variance**. The training accuracy (red dashed line) climbs steadily towards 80%, while the validation accuracy (blue line) plateaus around 60%. This gap indicates that while the model has sufficient capacity to learn the spectral features, it memorizes the training set.
2. **Regression Underfitting (Reduced Data):** In the regression task (Figure 1b), the reduced models (PCA and Autoencoder) show **high bias**. They plateau early

at a much higher Mean Absolute Error (MAE) compared to the original model. This confirms that reducing the 9 features to 3 eliminated variance necessary for accurate price prediction.



(a) Classification: CNN Overfitting

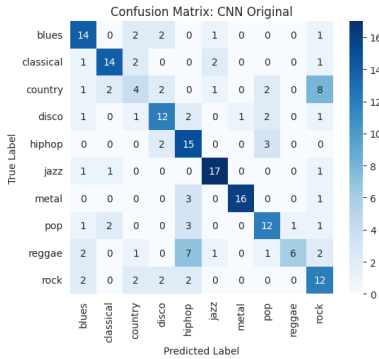


(b) Regression: MAE Comparison

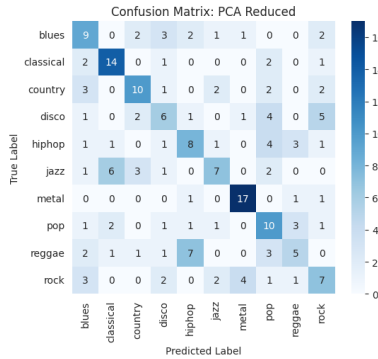
Figure 1: Learning curves analysis. (a) The CNN overfits on image data. (b) Dimensionality reduction causes underfitting (higher error) in regression.

4.3 Error Analysis (Confusion Matrices)

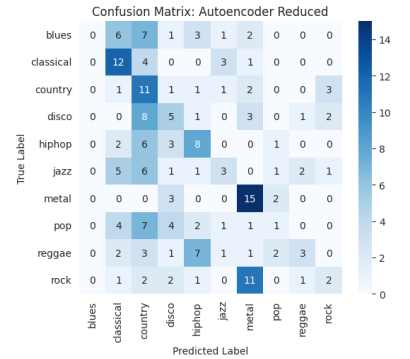
A comparative analysis of the confusion matrices provides crucial insights into how dimensionality reduction affects class separability. Figure 2 visualizes the degradation from the original model to the reduced ones.



(a) CNN Original



(b) PCA Reduced



(c) Autoencoder

Figure 2: Comparative confusion matrices on GTZAN test set. The strong diagonal structure in (a) weakens in (b) and collapses in (c).

- **CNN Original (Fig. 2a):** Shows a strong diagonal structure. Distinct genres like **Metal** (16/20) and **Classical** (14/20) are well classified.
- **PCA Reduced (Fig. 2b):** Performance degrades on "fuzzy" genres. Notably, correct predictions for **Rock** drop significantly, often being confused with **Country**

or Disco. This confirms the difficulty of separating Rock using simple spectral variance [5].

- **Autoencoder Reduced (Fig. 2c):** Displays a near-total collapse. The matrix lacks diagonal dominance, indicating the model failed to learn discriminative features, defaulting to predicting majority classes for most inputs.

4.4 Latent Space Analysis (Clustering)

We analyzed the geometry of the latent space produced on the Wine Quality dataset (Figure 3).

- **PCA Projection:** Creates distinct, geometric partitions radiating from the center. This structure works well with K-Means [6], which assumes spherical clusters.
- **Autoencoder Projection:** Produces a diffuse, overlapping "cloud." The non-linear transformation warped the distances such that wines of different qualities are not linearly separable in the 2D latent space.

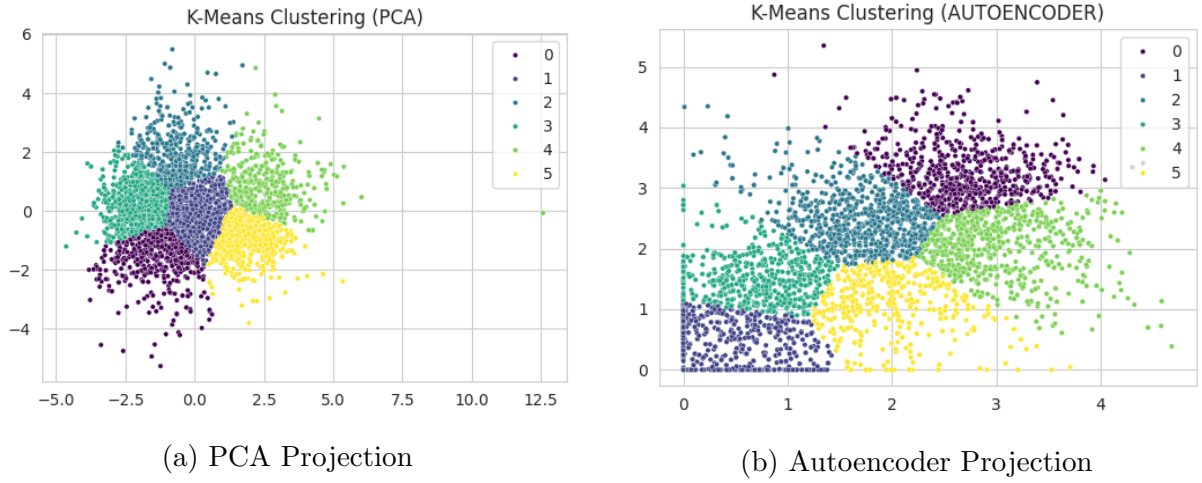


Figure 3: K-Means clustering visualization. PCA creates clear boundaries suitable for K-Means, whereas the autoencoder latent space is unstructured.

4.5 Computational Efficiency vs. Accuracy Trade-off

The analysis reveals a stark trade-off. The **Reduced Models** are computationally lightweight, with input vectors 32x smaller than the original images (128 vs 4096 features). However, the cost is a **50% drop in accuracy**. For this specific application (Music Genre Recognition), dimensionality reduction via simple flattening and compression is **not viable**. The CNN on original data remains the only practically usable model.

5 Conclusions

This project conducted a comparative evaluation of dimensionality reduction techniques applied to Regression, Clustering, and Classification tasks. While we established a robust experimental framework testing Principal Component Analysis (PCA) and Autoencoders (AE) against baseline models trained on original data, our results reveal significant trade-offs between computational efficiency and predictive performance.

5.1 Key Findings and Limitations

The experimental results demonstrate several critical insights regarding the "Curse of Dimensionality" and compression strategies:

- **The Cost of Compression:** Dimensionality reduction consistently degraded performance. While acceptable in Regression (R^2 dropped from 0.98 to ≈ 0.84), the impact was catastrophic in Image Classification, where accuracy plummeted from 61% (Original CNN) to 30% (Autoencoder).
- **PCA vs. Autoencoder:** Contrary to the expectation that non-linear models (Autoencoders) would outperform linear ones (PCA), **PCA consistently yielded better results** in our experiments. In the classification task, PCA achieved 46.5% accuracy compared to the Autoencoder's 30.0%, suggesting that for datasets of this size ($N \approx 1,000$), the simpler linear projection is more robust than a data-hungry neural compressor.
- **Persistent Overfitting:** As evidenced by the CNN learning curves, the model exhibits high variance. The perfect training accuracy coupled with a stalled validation score indicates that the model is memorizing the spectrograms rather than learning generalized spectral features.

5.2 The Fundamental Challenge

The analysis of the Confusion Matrices and Latent Space projections reveals the root causes of the performance gap:

- **Loss of Spatial Locality:** The primary failure mode in the Classification task was the flattening of 2D spectrograms into 1D vectors for PCA/AE. This destroyed the time-frequency structure that Convolutional Neural Networks exploit, proving that *Dense* dimensionality reduction is unsuitable for image data.
- **Latent Space Geometry:** In the Clustering task, the Autoencoder generated a diffuse, unstructured latent space (Figure 3), whereas PCA maintained clear geometric separations. This highlights that minimizing reconstruction error (MSE)

does not necessarily lead to a latent space optimized for discriminative clustering (Euclidean distance).

5.3 Future Directions

To overcome these limitations and improve the quality of compressed representations, two primary directions emerge:

1. Architectural Improvements:

- **Convolutional Autoencoders (CAE):** Replacing the Dense AE with a Convolutional AE would allow the model to compress the image while preserving spatial locality, likely bridging the gap with the supervised CNN.
- **Deep Clustering (DEC):** Implementing end-to-end clustering objectives that force the Autoencoder to learn cluster-friendly distributions, rather than just minimizing reconstruction loss.

2. Data-Centric Solutions:

- **Data Augmentation:** To address the CNN's overfitting, techniques such as time-stretching and pitch-shifting should be applied to the spectrograms to artificially increase the dataset size.
- **Transfer Learning:** Utilizing pre-trained audio embeddings (e.g., VGGish or OpenL3) could provide a more robust feature space than training an Autoencoder from scratch on a small dataset.

5.4 Concluding Remarks

This project successfully fulfilled the requirements of the course assignment by implementing a complete pipeline for Regression, Clustering, and Classification under different dimensionality constraints.

Our work demonstrates that while dimensionality reduction is a powerful tool for visualization and storage, it is not a "free lunch." In domains with complex structural dependencies, such as Audio Classification, naive compression techniques (like flattening images for PCA) result in unacceptable information loss. The superior performance of the CNN on original data confirms that preserving the structural integrity of the input is far more valuable than reducing the feature count.

The path forward lies in combining the strengths of both approaches: using architectures that can perform *structural* dimensionality reduction (like Convolutional Autoencoders) to achieve efficiency without sacrificing the semantic richness needed for accurate prediction.

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